

1 Notation & Perturbation Method

Big $\mathcal{O}$: $f(z) = \mathcal{O}(g(z))$ as $z \rightarrow z_0$ if $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} \text{bdd} \Leftrightarrow |f(z)| \leq A|g(z)|$ as $z \rightarrow z_0$

Small o : $f(z) = o(g(z))$ as $z \rightarrow z_0$ if $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = 0 \Leftrightarrow f(z) \ll g(z)$ as $z \rightarrow z_0$

Asymptotic $\sim$: $f(z) \sim g(z)$ as $z \rightarrow z_0$ if $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = 1$ **ps:** $e^{-x} = o(x^{-n}), \forall n > 0$ as $x \rightarrow \infty$

Asymptotic Sequence: Seq $\{\varphi_n(z)\}$ is asy. if $\forall n \in \mathbb{N} \varphi_{n+1}(z) = o(\varphi_n(z))$ as $z \rightarrow z_0$

Asymptotic Expansion: $f(z) \sim \sum_{n=0}^M a_n \varphi_n(z)$ as $z \rightarrow z_0$ to M if $\phi_n(z)$ is asy. seq. and

$f(z) - \sum_{n=0}^N a_n \varphi_n(z) = o(\varphi_N(z)) = \mathcal{O}(\varphi_{N+1}(z)), \forall N \leq M$ as $z \rightarrow z_0$

$f(x, \varepsilon) \sim \sum_{n=0}^{\infty} a_n(x) \varepsilon^n$ uniformly in $x \in I$ as $\varepsilon \rightarrow 0$ if ε 与 x 无关 (i.e. sup $|f(x, \varepsilon) - \sum_{n=0}^N a_n(x) \varepsilon^n| = o(\varepsilon^N)$)

Singular **Regular - Perturbation Method:** a. Let $\varepsilon = 0$ solve $f(x, 0) = 0$.

b.i. 如果 f 结构变了 (阶数降低), Let $x = \varepsilon^{-\alpha} X$, PDB 找 α c. 假设 $X = a_0 + a_1 \varepsilon^\beta + a_2 \varepsilon^{2\beta} + \dots$ 代入求解

b.ii. 如果 f 结构不变/变, 但有重根 x_0 , Let $x = x_0 + \varepsilon^\alpha X$, PDB 找 α . $\uparrow \beta$ 看化简后的 X 方程得出

Principle of dominant balance (PDB): For Singular Perturbation $p(x, \varepsilon) = 0$, Let $x = \varepsilon^\alpha X$, 代入选择 α :

对于 ε 来说 a. 至少有两项阶数相同且最低; b. 其他项阶数均更高. (即可忽略)

2 Integral By Parts, Laplace's Method & Method of Steepest Descents

Taylor: $f(x) = \sum_{n=0}^{N-1} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n + \frac{f^{(N)}(\xi)}{N!} (x - x_0)^N \Rightarrow f(x) \sim \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$ as $x \rightarrow x_0$

Exp. Integral: $E_1(x) = \int_x^\infty \frac{e^{-t}}{t} dt$, $(x \rightarrow 0 \text{ 发散})$ $E_1(x) \sim e^{-x} \sum_{n=0}^{\infty} \frac{(-1)^n}{x^{n+1}}, x \rightarrow \infty$

Gamma Func: $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$ $\Gamma(x+1) = x\Gamma(x)$ and $\Gamma(n) = (n-1)!$ $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

Incomplete Gamma Func: **Lower:** $\gamma(a, x) = \int_0^x t^{a-1} e^{-t} dt$ **Upper:** $\Gamma(a, x) = \int_x^\infty t^{a-1} e^{-t} dt$

$\gamma(a, x) = \Gamma(a) - \Gamma(a, x)$ $\Gamma(a, x) \sim e^{-x} x^{a-1} \sum_{n=0}^{\infty} \frac{\Gamma(a)}{\Gamma(a-n)} x^{-n}, x \rightarrow \infty$

Laplace's Method: For $I(x) = \int_a^b e^{x f(t)} g(t) dt$ as $x \rightarrow \infty$ $\frac{f''(t_0)}{2} (t - t_0)^2 + \dots$ $g(t_0) dt$

1. If $f(t)$ has unique max at $t = t_0 \in (a, b) \Rightarrow I \sim \int_{-\infty}^{\infty} e^{x f(t_0) + \frac{f''(t_0)}{2} (t - t_0)^2 + \dots} g(t_0) dt$

(留 n 项非常数展开为 n term approximation) (ps: $\max \leftrightarrow f'(t_0) = 0, f''(t_0) < 0$)

Useful Tool: Now: $I \sim e^{x f(t_0)} \int_{-\infty}^{\infty} e^{x [\frac{f''(t_0)}{2} (t - t_0)^2 + \dots]} g(t_0) dt \Rightarrow \text{Let } \tau^2 = -[\frac{1}{2} f''(t_0) (t - t_0)^2 + \dots]$

Assume $t = \tau + b\tau^2 + c\tau^3 + \dots$ as $\tau \rightarrow 0 \Rightarrow$ 代入换元式子求系数 $\Rightarrow dt = (1 + 2b\tau + 3c\tau^2 + \dots) d\tau$

2. **Exact Substitution:** 由 1 的基础, 得到 $f(t)$ 在 t_0 处的展开式, 换元令 $f(t)$ 就等于展开式前几项求解.

3. If $g = g(t, x): e^{x f(t)} g(t, x) = e^{x f(t) + \ln(g(t, x))}$, 看 $x f(t) + \ln(g(t, x))$ 的最大值, 类似 1 处理.

- If $t_0 = f(x)$, 换元令 $t = f(x)s$.

Method of Steepest Descents: For $I(x) = \int_C e^{x f(z)} g(z) dz$ as $x \rightarrow \infty$. f 容易避开 singular point

If z_0 is saddle point: $f'(z_0) = 0, f''(z_0) \neq 0$ $f(z)$ Taylor at z_0 : $f(z) = f(z_0) + \frac{f''(z_0)}{2} (z - z_0)^2 + \dots$

1. Let $z = z_0 + e^{i\theta} y$ (θ 选让 $f''(z_0) e^{2i\theta}$ 为负实数) $\rightarrow I(x) \sim \int_{-\infty}^{\infty} e^{x [f(z_0) - \frac{|f''(z_0)|}{2} y^2 + \dots]} g(z_0) e^{i\theta} dy$

2. If f 不容易避开 singular point: a. $\text{Im}(f(z)) = \text{Im}(f(z_0))$ and $z_0 \in C_0$ to find steepest descent path C_0

3. b. 画图看哪些 singular points 在 $C \rightarrow C_0$ 变形路径上, 计算 residues. c. $I = 1$ 部分 + $2\pi i \sum$ residues.

4. **More than one saddle point:** i. 如果通过积分区域可以舍弃, 则舍弃 ii. saddle points: z_k , 取能让 $\text{Re}(f(z_k))$ 最小的. iii. 如果多个 saddle points 都满足 $\text{Re}(f(z_k))$ 最小, 则都要计算后叠加.

5. **Exact Substitution:** 由 1 的基础, 得到 $f(z)$ 在 z_0 处的展开式, 换元令 $f(z)$ 就等于展开式前几项求解.

6. **Remark:** 类似 Laplace 3 一样, 可以对 $x f(x)$ 一个整体处理, 必要时 $z_0 = f(x) \rightarrow$ 换元 $z = f(x)s$.

Residue: If z_0 pole of order m of f , $\Rightarrow \text{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)]$

Method of Stationary Phase: $\cos(x f(\theta)) \rightarrow \text{Re}(e^{i x f(\theta)}) \rightarrow$ 回到 Steepest Descents, 同理 $\sin(x f(\theta))$

3 Perturbation Methods in ODEs

Regular Perturbation: For $L(y, \varepsilon) = 0$, let $y(x, \varepsilon) = y_0(x) + \varepsilon y_1(x) + \varepsilon^2 y_2(x) + \dots$

$\Rightarrow$ a. 代入 $\mathcal{O}(\varepsilon^n) \rightarrow$ b. Boundary/Initial Condition 求解 $y_n(x) \rightarrow$ a.

Singular Perturbation: e.g. $\varepsilon y'' + p(x)y' + q(x)y = 0, y(a) = A, y(b) = B, a < b$, 大致步骤:

1. Assume boundary layer at $x = a|b$. Let $x = a + \delta(\varepsilon) X$ or $x = b - \delta(\varepsilon) X$

2. Assume inner $Y(X) = y(x)$, chain rule 求导代入原 ODE $\rightarrow$ Domain balance 找出 $\delta(\varepsilon) = \varepsilon^\alpha$

Recall PDB: a. 至少有两项阶数相同且最低; b. 其他项阶数均更高.

3. Assume $Y = Y_0(X) + \varepsilon^\beta Y_1(X) + \varepsilon^{2\beta} Y_2(X) + \dots$ 代入求解 $\rightarrow$ 边界/初始条件求解 $Y_0(X)$

4. **Matching:** Consider $X \rightarrow \infty | x \rightarrow a|b$, If limits not exist (除非 X) $\rightarrow$ boundary layer 在另一个端点 $\rightarrow 1$

If limits exist, boundary layer 正确 $\rightarrow$ 渐进进入 (x), 算 inner 极限 (X); 算不了保留 $X \rightarrow x$ 和 X 的关系代入 $\rightarrow$ 比较得系数.

Remark 1: Outer solution 只能用在 away from boundary layer 的区域 (包括边界点).

Remark 2: Inner solution 只能用在 boundary layer 附近的区域 (包括边界点).

Useful Lemma: For $\varepsilon y'' + p(x)y' + q(x)y = 0, y(a) = A, y(b) = B, a < b, p(x) \neq 0$ in $[a, b]$ then

Has boundary layer at $x = a$ if $p > 0$; Has boundary layer at $x = b$ if $p < 0$.

4 WKB Theory & Solving PDE

WKB: For $u'' + k^2 q(x)u = 0, k \gg 1$ (or $k^2 q \gg 1$) we have: $u \sim A(x)e^{ik\phi(x)}$. Substitution can get:

If $q(x) < 0: u(x) \sim A_+ |q|^{-\frac{1}{4}} e^{-k \int_{x_0}^x \sqrt{|q(t)|} dt} + A_- |q|^{-\frac{1}{4}} e^{-k \int_{x_0}^x \sqrt{|q(t)|} dt}$ (Exp. Decay/Growth)

If $q(x) > 0: u(x) \sim A_+ q^{-\frac{1}{4}} e^{ik \int_{x_0}^x \sqrt{q(t)} dt} + A_- q^{-\frac{1}{4}} e^{-ik \int_{x_0}^x \sqrt{q(t)} dt}$ (Oscillation)

Turning Point: If $q(x_0) = 0$, then x_0 is a WKB turning point.

5 Appendix

5.1 ODE

$y' + P y = Q: \mu y = \int Q(x) \mu dx + C, \mu(x) = e^{\int P(x) dx}$

$y'' + k^2 y = 0: y = A \cos(kx) + B \sin(kx) \quad y'' - k^2 y = 0: y = A e^{kx} + B e^{-kx}, k > 0$

5.2 Integral

Gaussian Integral: $\int_{-\infty}^{\infty} e^{-ax^2} dx = \frac{\sqrt{\pi}}{\sqrt{a}} \quad \int_0^\infty e^{-xt^n} t^m = \frac{1}{n} \Gamma(\frac{m+1}{n}) x^{-\frac{m+1}{n}}$

Integral: $\int \tan x = -\ln |\cos x| = \ln |\sec x| \quad \int \cot x = \ln |\sin x| = -\ln |\csc x|$

Techniques: a. e.g. $\int e^{at} \cos(bt) dt = \text{Re}(\int e^{(a+ib)t} dt)$

5.3 Expansion

Binomial Expansion: $(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \dots$ as $x \rightarrow 0$

Taylor $[x \rightarrow 0]: f(x+h) \sim f(x) + f'(x)h + \frac{f''(x)}{2!} h^2 + \dots$ as $h \rightarrow 0$

$\sin(x) \sim x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$ as $x \rightarrow 0 \quad e^x \sim 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ as $x \rightarrow 0$

$\cos(x) \sim 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$ as $x \rightarrow 0 \quad \ln(1+x) \sim x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$ as $x \rightarrow 0$

$\sinh(x) \sim x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$ as $x \rightarrow 0 \quad \cosh(x) \sim 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$ as $x \rightarrow 0$

$\tan(x) \sim x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$ as $x \rightarrow 0 \quad \cot(x) \sim \frac{1}{x} - \frac{x}{3} - \frac{x^3}{45} - \dots$ as $x \rightarrow 0$

For $x \rightarrow \infty: \arctan(x) \sim \frac{\pi}{2} - \frac{1}{x} + \frac{1}{3x^3} - \dots$ as $x \rightarrow \infty$

$\ln(a+x) = \ln x + \ln(1 + \frac{a}{x}) \sim \ln x + \frac{a}{x} - \frac{a^2}{2x^2} + \dots$ as $x \rightarrow \infty$

$\frac{1}{a+x} = \frac{1}{x} \cdot \frac{1}{1+\frac{a}{x}} \sim \frac{1}{x} - \frac{a}{x^2} + \frac{a^2}{x^3} - \dots$ as $x \rightarrow \infty$

sin|cos: $a \sin(x) + b \cos(x) = \sqrt{a^2 + b^2} \sin(x + \varphi) \quad a \cos(x) + b \sin(x) = \sqrt{a^2 + b^2} \cos(x - \varphi), \varphi$ 过 (a, b) .

$-\sin(x) + \sin(y) = 2 \sin(\frac{x+y}{2}) \cos(\frac{x-y}{2}) \mid \sin(x) \cos(y) = \frac{1}{2} [\sin(x+y) + \sin(x-y)] \quad S+S=CS, S-S=CS, C+C=CC, S-S=-SS$

Inequalities: $\ln(x+1) < x, \quad e^x > x+1, \quad (1+x)^a < e^{ax}$

5.4 Other

Nabla: $\nabla^2 u = u_{xx} + u_{yy} + u_{zz} + \dots \quad \nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$